Improving Search Agent with One Line of Code

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Abstract

Retrieval-Augmented Generation (RAG), which integrates external knowledge, has become a standard approach to mitigate hallucinations and factual inaccuracies in Large Language Models (LLMs). Recently, search agents have reframed the RAG workflow as an autonomous multi-turn information-seeking process driven by Tool-based Agentic Reinforcement Learning (TARL). However, the Group Relative Policy Optimization (GRPO) employed in TARL introduces Importance Sampling Distribution Drift (ISDD). This drift often causes a sharp decline in the importance sampling ratio, resulting in training instabilities that manifest as catastrophic and irreversible model collapse. Although the clipped surrogate objective attempts to mitigate this issue through hard clipping, it fails to account for the distributional drift that accumulates with increased response lengths or search turns. In this paper, we propose **S**earch **A**gent **P**olicy **O**ptimization (**SAPO**) to enhance training stability. We analyze the phenomenon of increasing clipping rates and decreasing rewards, attributing these effects to ISDD. To address this, SAPO introduces a simple penalty term to enforce a token-level constraint on the distributional divergence between the current and old policies. Extensive experiments on single-hop and multi-hop QA benchmarks demonstrate that SAPO significantly outperforms existing methods, achieving an absolute performance improvement of 10.6 points (a relative gain of 31.5%) over the Search-R1 baseline.

1 Introduction

Searching for information Case & Given (2016) is a fundamental activity in daily life. Traditional search engines, such as Google and Bing, facilitate efficient access to relevant content by returning ranked web pages in response to user queries. However, as queries grow in complexity and users expectations for precision rise, traditional information retrieval techniques Kobayashi & Takeda (2000) often struggle to capture nuanced user intent, frequently yielding results that lack contextual relevance. Large language models (LLMs) Zhao et al. (2023) have demonstrated remarkable capabilities in natural language understanding, reasoning, and information synthesis. Nevertheless, they lack inherent access to real-time external knowledge and are prone to generating factual errors. Retrieval-augmented generation (RAG) Lewis et al. (2020) synergizes the strengths of LLMs and search engines to mitigate these limitations by conditioning generation on retrieved external passages. Yet, static RAG pipelines restrict the LLM's ability to autonomously determine when and what to search, thereby constraining flexibility in real-world applications.

Recent studies Li et al. (2025a) have introduced LLM-based search agents Guo et al. (2024) capable of interpreting user intent, planning search strategies, executing multi-turn search actions, and accumulating retrieved evidence for answer generation. Representative methods, such as Search-R1 Jin et al. (2025), employ Group Relative Policy Optimization (GRPO) Kaelbling et al. (1996); Shao et al. (2024) with rule-based rewards to train agents to interact with a search tool over large corpora. Consequently, GRPO as a Tool-based Agentic Reinforcement Learning (TARL) method has emerged as a widely adopted post-training algorithm for search agents, inspiring numerous follow-up works Mei et al. (2025); Shi et al. (2025); Qian & Liu (2025); Zhang et al. (2025b).

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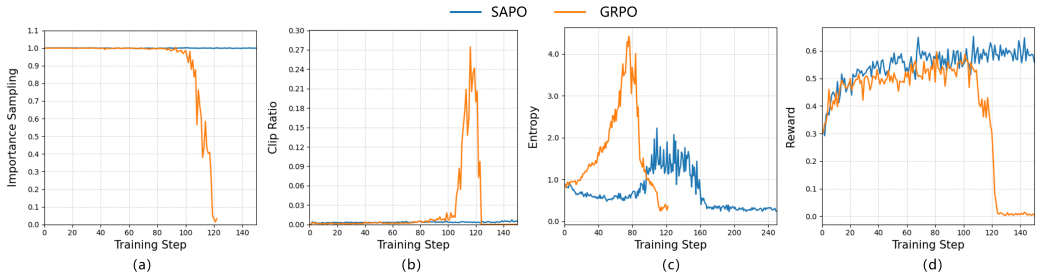

Figure 1: Comparison of training dynamics between SAPO and GRPO regarding (a) Importance Sampling Ratio, (b) Clip Ratio, (c) Entropy, and (d) Reward.

Despite promising results, existing TARL frameworks for search agents face a critical limitation: **the GRPO employed in TARL introduces Importance Sampling Distribution Drift (ISDD) outside the trust region.** Recently, several group-based optimization methods (Yu et al., 2025a; Zeng et al., 2025; He et al., 2025; Chen et al., 2025a; An et al., 2025; Wang et al., 2025a; Zhang et al., 2025a) have been proposed to adopt PPO_clip Schulman et al. (2017) for restricting the deviation of the current and old policies to address post-training RL instabilities. However, ISDD remains a common phenomenon in search agent training, particularly as response lengths or search turns increase. In GRPO, all tokens within a response share the same advantage value. which may suffer from two issues: (1) the advantage value may be numerically inaccurate, as different tokens contribute unequally to the final answer's correctness, and (2) The update direction could be misleading, as a response with a correct final answer might still include incorrect intermediate steps or a response with an incorrect final answer might still include correct intermediate steps. Conflicts arise between the current and old policies when the old policies think some tokens are correct, but the current policies disagree due to negative advantages. As shown in Figure. 1, this conflict often causes a sharp decline in the importance sampling (IS) ratio, resulting in training instabilities.

To enhance training stability, we propose **S**earch **A**gent **P**olicy **O**ptimization (SAPO), a simple yet theoretically grounded modification of GRPO that alleviates the conflict between the current and old policies. Following PPO_KL Schulman et al. (2017), SAPO introduces a KL penalty term to enforce a token-level constraint on the distributional divergence between the current and old policies. To optimize the constraint efficiency, SAPO employs an asymmetric constraint for positive and negative tokens, thereby preserving IS weights for positive tokens. In addition, SAPO selectively constrains conflicting tokens while retaining informative gradients from positive tokens with lower IS ratios. This selective constraint mitigates the signal loss associated with hard clipping.

We summarize the contributions as follows.

- We propose SAPO, a search agent policy optimization method that improves post-training stabilization for autonomous multi-turn information-seeking agents addressing complex, real-world questions.
- We introduce a selective KL penalty term to enforce a token-level constraint on the distributional divergence between the current and old policies.
- We demonstrate the effectiveness and generalizability of SAPO across different LLM-based search agents on seven diverse and challenging QA benchmarks.

2 Preliminaries

2.1 Proximal Policy Optimization

Proximal Policy Optimization (PPO) Schulman et al. (2017) stands as one of the most prominent policy gradient methods in reinforcement learning (RL) Kaelbling et al. (1996). PPO aims to stabilize training by adopting IS to constrain the deviation between the current

and old policies during updates, thereby preventing excessively large policy shifts. The standard unclipped PPO surrogate objective is defined as:

$$\mathcal{J}_{\text{PG}}(\theta) = \mathbb{E}_{q\sim\mathcal{D},\, o\sim\pi_{\theta_{\text{old}}}(\cdot|q)}\left[r_t(\theta)A_t\right], \quad \text{where} \quad r_t(\theta) = \frac{\pi_\theta(o_t \mid x, o_{<t})}{\pi_{\theta_{\text{old}}}(o_t \mid x, o_{<t})} \tag{1}$$

Here, π_{old} and π_θ denote the old policy and the current policy, respectively, while A_t represents the advantage function. q denotes a query sampled from the data distribution $\mathcal{D}$, and o represents a response generated by the old policy π_{old}. Direct optimization of this objective can lead to excessive deviation in the IS ratio $r_t(\theta)$, resulting in training instability. To mitigate this, PPO introduces a clipped surrogate objective that limits the IS ratio to a defined interval, effectively imposing a local trust region:

$$\mathcal{J}_{\text{PPO}}(\theta) = \mathbb{E}_{q\sim\mathcal{D},\, o\sim\pi_{\theta_{\text{old}}}(\cdot|q)}\left[\min\left(r_t(\theta)A_t,\, \text{clip}(r_t(\theta), 1-\epsilon, 1+\epsilon)\, A_t\right)\right] \tag{2}$$

Here, ϵ is a hyperparameter defining the clipping bounds. This mechanism limits the impact of updates with large IS ratios, thereby reducing variance and enhancing stability. However, this clipping mechanism clips the value and also masks the gradient of the clipped tokens, which may still drift beyond the trust region.

2.2 Group Relative Policy Optimization

Group Relative Policy Optimization (GRPO) Guo et al. (2025) employs a group of rollouts for advantage estimation. In search agents, the policy model π_θ interacts with search engines and samples an interleaved sequence $o_0, k_0, o_1, k_1, \ldots, o_T, k_T$. Following Search-R1 Jin et al. (2025), external retrieval tokens k are masked during loss computation. The GRPO token-level objective is optimized solely based on the model outputs o:

$$\mathcal{J}_{\text{GRPO}}(\theta) = \mathbb{E}_{q\sim\mathcal{D},\{o_i\}_{i=1}^{G}\sim\pi_{\theta_{\text{old}}}(\cdot|q)}\left[\frac{1}{G}\sum_{i=1}^{G}\frac{1}{|o_i|}\sum_{t=1}^{|o_i|}\min\left(r_{i,t}(\theta)\hat{A}_{i,t}, \text{clip}\left(r_{i,t}, 1-\epsilon, 1+\epsilon\right)\hat{A}_{i,t}\right)\right] \tag{3}$$

where

$$r_{i,t}(\theta) = \frac{\pi_\theta(o_{i,t} \mid q, o_{i,<t})}{\pi_{\theta_{\text{old}}}(o_{i,t} \mid q, o_{i,<t})}, \qquad \hat{A}_{i,t} = \hat{A}_i = \frac{R_i - \text{mean}(\{R_j\}_{j=1}^{G})}{\text{std}(\{R_j\}_{j=1}^{G}))} \tag{4}$$

For each query, a group of G outputs is sampled. The advantages are estimated by normalizing the corresponding sequence rewards R across this group

3 Approach

3.1 Search Agent

Let q denote the user question, a the generated answer, and π_θ the LLM. We model the standard question-answering task as

$$a = \pi_\theta(q) \tag{5}$$

RAG systems enhance this process by retrieving knowledge k from an external corpus $\mathcal{D}$ using a retriever $\mathcal{R}(\cdot)$, subsequently generating an answer conditioned on both q and k.

$$a = \pi_\theta(q \mid k), \quad k = \mathcal{R}(q \mid \mathcal{D}) \tag{6}$$

However, this static workflow is often insufficient for complex knowledge discovery tasks that necessitate iterative reasoning and adaptive retrieval. Recent advancements in search agents enable a more dynamic paradigm in which retrieval is interleaved with reasoning. In this setting, the model output o comprises intermediate reasoning steps r and the final answer a.

$$a = \pi_\theta \left(q \mid \{r_t \otimes k_t\}_{t=1}^{T} \right), k_t = \mathcal{R}(sq_t \mid \mathcal{D}), sq_t \in r_t \tag{7}$$

Here, sq_t denotes a search query generated within the reasoning step r_t to retrieve intermediate knowledge k_t, and the operator $\otimes$ indicates that the retrieved knowledge is integrated into the reasoning trajectory. To eliminate the need for a separate value estimator, we adopt GRPO as our TARL framework.

3.2 Importance Sampling Distribution Drift

The The application of GRPO in search agents often includes Importance Sampling Distribution Drift (ISDD). This phenomenon causes training instabilities, frequently manifesting as catastrophic and irreversible model collapse. Figure.1 illustrates the training dynamics of Search-R1 using GRPO, specifically tracking the IS ratio, clip ratio, entropy, and reward. While the IS ratio remains near 1 during the initial training phase, it subsequently drops sharply, indicating a growing divergence between the current and old policies. Consequently, the clip ratio exhibits a pronounced spike in the latter stages of training. Similarly, policy entropy rises substantially at the beginning but collapses abruptly thereafter. The outcome-based reward initially increases but degrades as the IS ratio destabilizes and entropy collapses, suggesting the late-stage instability harms final performance. These empirical results demonstrate that despite the use of hard clipping on the IS ratio, catastrophic and irreversible model collapse persists.

The primary motivation behind IS and clipping is to correct for distribution shifts and ensure training stability. However, for the positive tokens (i.e., actions with positive advantages), when the probability under the current policy drops significantly below that of the old policy, the update is effectively suppressed by excessively low importance weights. This yields a weak update signal, leading to insufficient learning. This observation prompts the following research question:

How can we mitigate optimization conflicts using appropriate constraints when the current policy diverges significantly from the old policy on positive tokens.

3.3 Search Agent Policy Optimization

To alleviate ISDD and further enhance training stability, SAPO introduces an auxiliary penalty term. This term enforces a global constraint on the distributional difference between the current and old policies. SAPO employs the following token-level optimization objective.

$$\begin{aligned} \mathcal{J}_{\text{SAPO}}(\theta) = & \mathbb{E}_{q \sim \mathcal{D}, \{o_i\}_{i=1}^{G} \sim \pi_{\theta_{\text{old}}}(\cdot|q)} \Bigg[\frac{1}{G} \sum_{i=1}^{G} \frac{1}{|o_i|} \sum_{t=1}^{|o_i|} \\ & \min \left(r_{i,t}(\theta) \hat{A}_{i,t}, \text{clip}\left(r_{i,t}, 1-\epsilon, 1+\epsilon\right) \hat{A}_{i,t} \right) - \gamma \text{KL}\left[\pi_\theta \parallel \pi_{\text{old}}\right] \Bigg] \end{aligned} \tag{8}$$

where γ denotes the KL penalty coefficient. Unlike the KL penalty in GRPO, which regulates the divergence between the current policy and a frozen reference policy, SAPO imposes a penalty on the divergence between the current policy and the dynamic old policy. Specifically, we approximate the KL divergence following PPO Schulman et al. (2017).

$$\text{KL}\left[\pi_\theta \parallel \pi_{\text{old}}\right] = \log \frac{\pi_\theta(o_{i,t} \mid q, o_{i,<t})}{\pi_{\theta_{\text{old}}}(o_{i,t} \mid q, o_{i,<t})} = \log r_{i,t}(\theta) \tag{9}$$

However, the standard KL divergence penalty used in PPO may hinder policy exploration. To address this, we propose a conditional KL penalty that activates only when the probability ratio decreases for samples with positive advantages.

$$\mathrm{KL}\left[\pi_\theta \,\|\, \pi_{\text{old}}\right] = \mathbb{I}(r_{i,t}(\theta) < 1, \hat{A}_{i,t} > 0) \log r_{i,t}(\theta) \tag{10}$$

where $\mathbb{I}(\cdot)$ is the indicator function. Then, the final SAPO objective is formulated as:

$$\begin{aligned}\mathcal{J}_{\text{SAPO}}(\theta) =& \mathbb{E}_{q\sim\mathcal{D},\{o_i\}_{i=1}^{G}\sim\pi_{\theta_{\text{old}}}(\cdot|q)}\Bigg[\frac{1}{G}\sum_{i=1}^{G}\frac{1}{|o_i|}\sum_{t=1}^{|o_i|} \\ & \min\Big(r_{i,t}(\theta)\hat{A}_{i,t}, \text{clip}\left(r_{i,t}, 1-\epsilon, 1+\epsilon\right)\hat{A}_{i,t}\Big) - \gamma\mathbb{I}(r_{i,t}(\theta) < 1, \hat{A}_{i,t} > 0)\log r_{i,t}(\theta)\Bigg]\end{aligned} \tag{11}$$

3.4 Gradient Analysis

We derive the gradient of the SAPO KL penalty term as follows:

$$\begin{aligned}\nabla_\theta \mathcal{J}_{\text{SAPO_KL}}(\theta) &= \nabla_\theta \mathbb{I}(r_{i,t}(\theta) < 1, \hat{A}_{i,t} > 0)\log r_{i,t}(\theta) \\ &= \mathbb{I}(\hat{A}_{i,t} > 0)\nabla_\theta\left[\mathbb{I}(r_{i,t}(\theta) < 1)\log r_{i,t}(\theta)\right] \\ &= \mathbb{I}(\hat{A}_{i,t} > 0)[\mathbb{I}(r_{i,t}(\theta) < 1)\nabla_\theta \log r_{i,t}(\theta) + \nabla_\theta \mathbb{I}(r_{i,t}(\theta) < 1)\log r_{i,t}(\theta)] \\ &= \mathbb{I}(\hat{A}_{i,t} > 0)[\mathbb{I}(r_{i,t}(\theta) < 1)\nabla_\theta \log r_{i,t}(\theta) + \delta(1 - r_{i,t}(\theta))\nabla_\theta r_{i,t}(\theta)\log r_{i,t}(\theta)] \\ &= [\mathbb{I}(r_{i,t}(\theta) < 1, \hat{A}_{i,t} > 0) + \delta(1 - r_{i,t}(\theta))r_{i,t}(\theta)\log r_{i,t}(\theta)]\nabla_\theta \log \pi_\theta(o_{i,t}|q, o_{i,<t})\end{aligned} \tag{12}$$

Subsequently, the total gradient of the SAPO objective is formulated as:

$$\begin{aligned}\nabla_\theta \mathcal{J}_{\text{SAPO}}(\theta) = \mathbb{E}_{q\sim\mathcal{D},\{o_i\}_{i=1}^{G}\sim\pi_{\theta_{\text{old}}}(\cdot|q)}\Bigg[\frac{1}{G}\sum_{i=1}^{G}\frac{1}{|o_i|}\sum_{t=1}^{|o_i|}(\hat{A}_{i,t}r_{i,t}(\theta) \\ -\gamma[\mathbb{I}(r_{i,t}(\theta) < 1, \hat{A}_{i,t} > 0) + \delta(1 - r_{i,t}(\theta))r_{i,t}(\theta)\log r_{i,t}(\theta)])\nabla_\theta \log \pi_\theta(o_{i,t}|q, o_{i,<t})\Bigg].\end{aligned} \tag{13}$$

where $\delta(\cdot)$ denotes the Dirac delta function, arising from the differentiation of the indicator function $\mathbb{I}(\cdot)$.

4 Experiments

4.1 Experiment Settings

Datasets. We evaluate our method on seven diverse QA benchmarks, categorized into single-hop or multi-hop retrieval tasks. The single-hop datasets include Natural Questions (NQ) Kwiatkowski et al. (2019), TriviaQA Joshi et al. (2017), and PopQA Mallen et al. (2022). The multi-hop datasets comprise HotpotQA Yang et al. (2018), 2WikiMultihopQA Ho et al. (2020), Musique Trivedi et al. (2022b), and Bamboogle Press et al. (2022). We employ Exact Match (EM) accuracy as the evaluation metric across all datasets. Following Search-R1 Jin et al. (2025), SAPO is trained on acomposite dataset consisting of NQ and HotpotQA.

Baselines. We compare SAPO against three categories of methods: (1) **Retrieval-free methods**: direct LLM generation, Supervised Fine-Tuning (SFT), and R1-style training Guo et al. (2025) without retrieval; (2) **Retrieval-enhanced workflows**: Naive RAG, which

Methods	Single-Hop QA (EM)				Multi-Hop QA (EM)					QA (EM)
	NQ	TriviaQA	PopQA	Avg.	HotpotQA	2Wiki	Musique	Bamboogle	Avg.	Avg.
w/o Retrieval										
Direct Generation	0.106	0.288	0.108	0.167	0.149	0.244	0.020	0.024	0.109	0.134
SFT	0.249	0.292	0.104	0.215	0.186	0.248	0.044	0.112	0.148	0.176
R1-Instruct Guo et al. (2025)	0.210	0.449	0.171	0.277	0.208	0.275	0.060	0.192	0.184	0.224
R1-Base Guo et al. (2025)	0.226	0.455	0.173	0.285	0.201	0.268	0.055	0.224	0.187	0.229
Workflow w/ Retrieval										
Naive RAG Lewis et al. (2020)	0.348	0.544	0.387	0.426	0.255	0.226	0.047	0.080	0.152	0.270
IRCoT Trivedi et al. (2022a)	0.111	0.312	0.200	0.208	0.164	0.171	0.067	0.240	0.161	0.181
Agent w/ Retrieval										
Search-o1 Li et al. (2025c)	0.238	0.472	0.262	0.324	0.221	0.218	0.054	0.320	0.203	0.255
Search-R1-Instruct Jin et al. (2025)	0.397	0.565	0.391	0.451	0.331	0.310	0.124	0.232	0.249	0.336
Search-R1-Base Jin et al. (2025)	0.421	0.583	0.413	0.472	0.297	0.274	0.066	0.128	0.191	0.312
ReSearch-Instruct Chen et al. (2025b)	0.365	0.571	0.395	0.444	0.351	0.272	0.095	0.266	0.246	0.331
ReSearch-Base Chen et al. (2025b)	0.427	0.597	0.430	0.485	0.305	0.272	0.074	0.128	0.195	0.319
ZeroSearch-Base Sun et al. (2025)	0.430	0.616	0.414	0.487	0.338	0.346	0.130	0.139	0.238	0.345
StepSearch-Base Wang et al. (2025b)	-	-	-	-	0.329	0.339	0.181	0.328	0.294	-
O^2-Searcher Mei et al. (2025)	0.444	0.597	0.429	0.490	0.388	0.374	0.160	0.344	0.317	0.391
AutoRefine-Instruct Shi et al. (2025)	0.436	0.597	0.447	0.493	0.404	0.380	0.169	0.336	0.322	0.396
AutoRefine-Base Shi et al. (2025)	0.467	0.620	0.450	0.512	0.405	0.393	0.157	0.344	0.325	0.405
InForage Qian & Liu (2025)	0.421	0.597	0.452	0.490	0.409	0.428	0.172	0.360	0.342	0.405
CriticSearch Zhang et al. (2025b)	-	-	-	-	0.414	0.409	0.180	0.368	0.343	-
SAPO-3B-Base	**0.474**	**0.630**	0.449	0.517	0.449	0.412	0.196	0.424	0.370	0.433
SAPO-3B-Instruct	0.469	0.629	**0.457**	**0.518**	**0.456**	**0.448**	**0.203**	**0.432**	**0.396**	**0.442**

Table 1: Accuracy comparison of SAPO-3B against baseline methods using Qwen2.5-3B Qwen et al. (2025) across multiple QA benchmarks. **Bold** indicates best results, and underline denotes second best results. Higher values denote better performance.

retrieves documents based solely on the input question, and IRCoT Trivedi et al. (2022a), which interleaves retrieval with Chain-of-Thought reasoning; (3) **Retrieval-augmented agents**: recent agentic methods including Search-o1 Li et al. (2025c), Search-R1 Jin et al. (2025), ReSearch Chen et al. (2025b), ZeroSearch Sun et al. (2025), StepSearch Wang et al. (2025b), O^2-Searcher Mei et al. (2025), InForage Qian & Liu (2025), CriticSearch Zhang et al. (2025b) and AutoRefine Shi et al. (2025).

Implementation Details. To simulate a realistic search environment, we utilize the December 2018 Wikipedia dump Karpukhin et al. (2020) as the external corpus, with E5-base-v2 Wang et al. (2022) serving as the retrieval engine. Unless otherwise specified, the retriever returns the top-3 documents per query. We employ the Qwen2.5 series Qwen et al. (2025) as the backbone language model. The maximum number of search turns is set to $T_{max} = 5$, and the KL penalty coefficient is set to $\gamma = 0.1$.

4.2 Main Performance

Table 1 presents the main experimental results comparing SAPO against baseline methods. The Avg. column reports the average accuracy. All methods utilize the same retriever, knowledge corpus, training data, and LLM backbone (Qwen2.5-3B).

SAPO outperforms other methods. SAPO surpasses Search-R1 and state-of-the-art methods (*i.e.*, AutoRefine and CriticSearch) across all seven benchmarks. Our method achieves an average EM accuracy of 0.442, demonstrating that the proposed SAPO post-training strategy substantially enhances the search agent's capabilities. Specifically, SAPO yields a significant absolute improvement of 10.6 points (a relative gain of 31.5%) over the Search-R1 baseline.

SAPO demonstrates particularly strong gains on multi-hop QA benchmarks. Performance improvements are most pronounced on complex, multi-hop QA tasks. For instance, compared to the CriticSearch, SAPO improves performance on HotpotQA by 4.2 percentage points (a 10.1% relative improvement) and on Bamboogle by 6.4 percentage points (a 17.4% relative improvement). Furthermore, across all multi-hop benchmarks, SAPO achieves an average increase of 5.3 percentage points (a 15.45% relative improvement). compared to Search-R1, SAPO achieves a substantial increase of 14.7 percentage points (a 24.9% relative improvement) across the four multi-hop QA benchmarks. These gains are attributed to the training stability provided by our proposed auxiliary KL penalty term.

Method	Single-Hop QA			Multi-Hop QA				Avg.
	NQ	TriviaQA	PopQA	HotpotQA	2Wiki	Musique	Bamboogle	
Search-R1-GRPO	0.402	0.662	0.436	0.374	0.376	0.134	0.333	0.388
Search-R1-GRPO	0.425	0.625	0.443	0.428	0.371	0.175	0.342	0.398
+ KL	+0.023	−0.037	+0.007	+0.054	−0.005	+0.041	+0.009	+0.010
	↑ 5.72%	↓ 5.59%	↑ 1.61%	↑ 14.44%	↓ 1.33%	↑ 30.60%	↑ 2.70%	↑ 2.58%
Search-R1-GRPO	0.439	0.660	0.456	**0.457**	0.369	0.169	0.369	0.417
+ KL_r	+0.014	+0.035	+0.013	+0.029	−0.002	−0.006	+0.027	+0.019
	↑ 3.29%	↑ 5.60%	↑ 2.93%	↑ 6.78%	↓ 0.54%	↓ 3.43%	↑ 7.89%	↑ 4.77%
Search-R1-GRPO	**0.442**	**0.673**	**0.474**	0.433	**0.415**	**0.181**	**0.382**	**0.429**
+ KL_{ra}	+0.003	+0.013	+0.018	−0.024	+0.046	+0.012	+0.013	+0.012
(SAPO)	↑ 0.68%	↑ 1.97%	↑ 3.95%	↓ 5.25%	↑ 12.47%	↑ 7.10%	↑ 3.52%	↑ 2.88%

Table 2: Ablation results showing the effect of SAPO. The first row reports the baseline method. Subsequent rows show performance after adding different KL terms.

Methods	Single-Hop QA			Multi-Hop QA				Avg.
	NQ	TriviaQA	PopQA	HotpotQA	2Wiki	Musique	Bamboogle	
Qwen2.5-1.5B-SAPO (EM)	0.440	0.591	0.432	0.413	0.381	0.160	0.384	0.400
Qwen2.5-1.5B-SAPO (F1)	0.528	0.671	0.480	0.532	0.445	0.247	0.487	0.484
Qwen2.5-3B-SAPO (EM)	0.469	0.629	0.457	0.456	0.448	0.203	0.432	0.442
Qwen2.5-3B-SAPO (F1)	0.562	0.708	0.504	0.581	0.517	0.299	0.553	0.533
Qwen2.5-7B-SAPO (EM)	0.467	0.648	0.449	0.453	0.432	0.214	0.504	0.452
Qwen2.5-7B-SAPO (F1)	0.559	0.723	0.493	0.579	0.506	0.314	0.609	0.540
Qwen2.5-14B-SAPO (EM)	0.510	0.695	0.485	0.494	0.477	0.255	0.548	0.495
Qwen2.5-14B-SAPO (F1)	0.602	0.770	0.527	0.621	0.549	0.353	0.680	0.586

Table 3: Generalization results showing the scaling trend across backbone LLMs (Qwen2.5-instruct) of sizes 1.5B, 3B, 7B, and 14B.

4.3 Ablation Studies

We conduct ablation study to evaluate the impact of SAPO's key components. To ensure a fair comparison, all models are trained for the same 300 steps, sampling 5 rollout responses. The baseline, Search-R1, utilizes Qwen2.5-3B-Instruct as the language model and the F1 score as the outcome-based reward. Table 2 details the accuracy across seven benchmarks and quantifies the contribution of each component. We compare four configurations: (1)**GRPO**: The baseline GRPO algorithm as employed in Search-R1. (2)**GRPO w/ KL**: GRPO augmented with an unconditional KL penalty term. (3)**GRPO w/ KL_r**: GRPO with a KL penalty conditioned solely on the IS ratio. (4)**GRPO w/ KL_ra (SAPO)**: The complete SAPO configuration, where the KL penalty is conditioned on both the IS ratio and the advantage.

KL penalty terms yield consistent improvements. Table.2 details the ablation study, starting with the Search-R1-GRPO baseline (average accuracy 0.388). Subsequent rows incrementally introduce the KL-based components. Subsequent rows incrementally introduce the KL-based components. Adding a simple KL term yields mixed gains on Musique (+30.6%) and HotpotQA (+14.4%) but small losses on TriviaQA and 2Wiki, resulting in a modest average increase to 0.398. The ratio-conditioned variant (KL_r) further improves stability and boosts HotpotQA to 0.457, producing an average of 0.417. Our full SAPO configuration (KL_ra) achieves the best overall balance, securing the highest scores on six out of seven benchmarks with an average of 0.429, an absolute increase of 0.041 over the baseline (about 10.6% relative). These results indicate that the selective, asymmetric KL penalties in SAPO produce consistent aggregate improvements while retaining the benefits of earlier variants.

4.4 Generalization and Scalability

Consistent scaling with model size. We investigate the scalability of SAPO by evaluating its performance across backbone models of varying sizes, specifically the Qwen2.5-Instruct series ranging from 1.5B to 14B parameters. To isolate the impact of model scale, we maintain a fixed configuration for the retriever, knowledge corpus, and evaluation protocols. Table 3 reports the Exact Match (EM) and F1 scores. The results exhibit a clear positive correlation between model size and performance, consistent with established scaling laws Kaplan et al.

Methods	Single-Hop QA			Multi-Hop QA				Avg.
	NQ	TriviaQA	PopQA	HotpotQA	2Wiki	Musique	Bamboogle	
LLaMA3.2-3b-Base								
Search-R1-GRPO	0.394	0.596	0.437	0.280	0.264	0.056	0.105	0.305
Search-R1-SAPO	0.471	0.630	0.452	0.355	0.301	0.102	0.248	0.366
LLaMA3.2-3b-Instruct								
Search-R1-GRPO	0.357	0.578	0.378	0.314	0.233	0.090	0.306	0.322
Search-R1-SAPO	0.475	0.634	0.446	0.444	0.386	0.187	0.384	0.422

Table 4: Generalization results on backbone LLMs (LLaMA3.2-3b).

(2020). As the parameter count increases, SAPO demonstrates a monotonic improvement in average accuracy. Notably, scaling from 1.5B to 14B results in a substantial performance leap, raising the average EM from 0.400 to 0.495 and the F1 score from 0.484 to 0.586. This confirms that SAPO effectively leverages the enhanced reasoning capabilities inherent in larger language models.

Robustness across model families. To verify that the effectiveness of SAPO is not limited to a specific architecture, we extend our evaluation to the LLaMA-3.2Touvron et al. (2023) series, testing both the Base and Instruct versions of the 3B model. As shown in Table 4, SAPO demonstrates consistent superiority over the GRPO baseline across both model variants. On LLaMA-3.2-3B-Base, SAPO improves the average accuracy from 0.305 to 0.366, proving its ability to effectively align even raw, pre-trained models. The gains are even more substantial on LLaMA-3.2-3B-Instruct, where SAPO boosts the average accuracy by 10.0 percentage points (from 0.322 to 0.422). These results confirm that SAPO is a model-agnostic framework capable of enhancing search capabilities regardless of the underlying model family or its prior alignment stage.

5 Related Works

5.1 Search Agents

Search agents augment Retrieval-Augmented Generation (RAG) Gao et al. (2023) by invoking a search tool within a multi-step reasoning process, as in Search-o1 Li et al. (2025c). Related systems such as WebThinker Li et al. (2025d) and WebDancer Wu et al. (2025) also support autonomous retrieval during reasoning. Other methods focus on training strategies that improve search behavior, including Search-R1 Jin et al. (2025), ReSearch Chen et al. (2025b), AutoRefine Shi et al. (2025), and WebSailor Li et al. (2025b). A common limitation among these methods is post-training instability, which often manifests as catastrophic and irreversible model collapse. To address this, this paper proposes Search Agent Policy Optimization (SAPO) to enhance training stability.

5.2 Reinforcement Learning

Reinforcement learning (RL) provides a general framework for improving LLM behavior. Common approaches include proximal policy optimization (PPO) Schulman et al. (2017), direct preference optimization (DPO) Rafailov et al. (2023), and group relative policy optimization (GRPO) Shao et al. (2024). GRPO specifically employs group-based normalization and eliminates the need for a separate critic model. Extensions such as dynamic sampling policy optimization (DAPO) Yu et al. (2025b) and group sequence policy optimization (GSPO) Zheng et al. (2025) further improve training stability and efficiency. In this work, we introduce a sample-based penalty term to enforce a token-level constraint on the distributional divergence between the current and old policies.

6 Conclusion

Tool-based agentic reinforcement learning, such as GRPO, introduces Importance Sampling Distribution Drift (ISDD). This drift often causes the importance sampling ratio to decline sharply, leading to training instabilities that manifest as catastrophic and irreversible model collapse. In this paper, we propose SAPO (**S**earch **A**gent **P**olicy **O**ptimization) to enhance the training stability of search agents. We analyze the phenomena of increasing clipping rates and decreasing rewards, attributing these effects to ISDD. To mitigate this, SAPO introduces a sample-based penalty term that enforces a token-level conditonal constraint on the distributional divergence between the current and old policies. Extensive experiments on single-hop and multi-hop QA benchmarks demonstrate that SAPO significantly outperforms existing methods, improving absolute performance by 10.6 points over the Search-R1 baseline.

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